

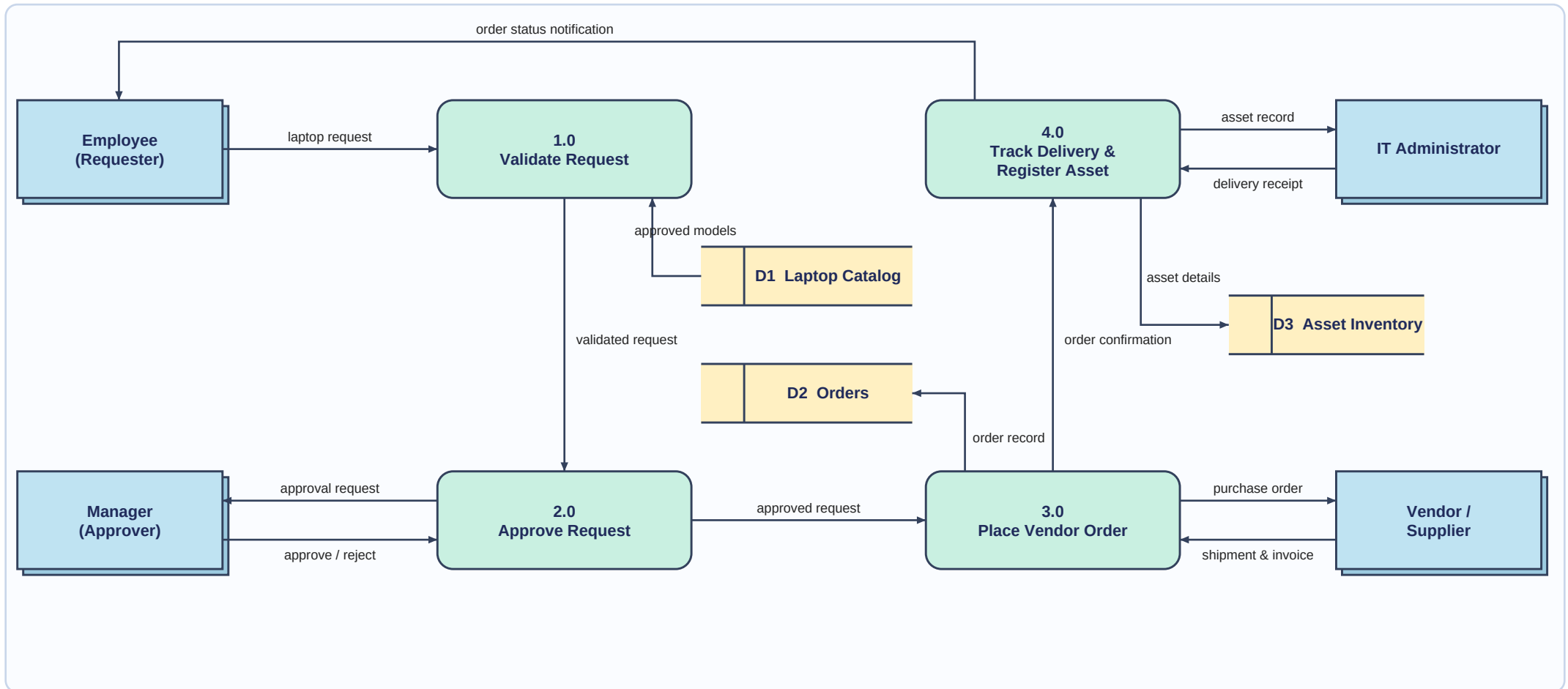
Project Design Phase-II

Data Flow Diagram & User Stories

Date	01 October 2026
Team ID	SWTID-2026-3408
Project Name	Streaming IT Procurement: Automating Standard Laptop Orders
Maximum Marks	4 Marks

Data Flow Diagram (Level 0)

The DFD shows how a standard laptop request enters the procurement system, is validated against the approved catalog, approved by a manager, ordered automatically from the vendor, and tracked until the laptop is delivered and registered as an asset.



Legend: Blue box = external entity | Green rounded box = process | Yellow open box = data store | Arrow = data flow

Data Flow Description

1	Employee submits a standard laptop request (model, quantity, delivery location) through the portal.
2	Process 1.0 checks the request against the approved models and prices held in D1 Laptop Catalog.
3	The validated request is forwarded to Process 2.0, which sends an approval request to the Manager.
4	Manager approves or rejects the request; the decision is returned to Process 2.0 and the employee is informed.
5	Approved requests go to Process 3.0, which automatically generates a purchase order and sends it to the Vendor.
6	Process 3.0 stores the order details in D2 Orders and receives the shipment details and invoice from the Vendor.
7	Order confirmation is passed to Process 4.0, which tracks delivery and sends status notifications to the Employee.
8	On arrival, the IT Administrator confirms the delivery receipt and the laptop is registered in D3 Asset Inventory.

User Stories

User stories for the Streaming IT Procurement system, grouped by user type and functional requirement (epic).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee (Requester)	Login	USN-1	As an employee, I can log in to the portal using my company email and password (SSO).	I can access my procurement dashboard.	High	Sprint-1
	Laptop Catalog	USN-2	As an employee, I can view the list of standard laptop models approved by IT, with specifications and prices.	I can see all approved models with specs and price.	High	Sprint-1
	Laptop Request	USN-3	As an employee, I can submit a standard laptop order by selecting a model, quantity and delivery location.	A request is created with a unique ID and status 'Submitted'.	High	Sprint-1
		USN-4	As an employee, I will receive a confirmation email once I submit my request.	I receive an email containing my request ID.	Medium	Sprint-1
	Order Tracking	USN-5	As an employee, I can track my order status (Submitted, Approved, Ordered, Shipped, Delivered).	Status updates are visible on my dashboard.	Medium	Sprint-2
Manager (Approver)	Approval	USN-6	As a manager, I will be notified when a team member's laptop request is waiting for approval.	I receive an email / in-app notification with an approval link.	High	Sprint-2
		USN-7	As a manager, I can approve or reject a request and add comments.	My decision is saved and the employee is notified.	High	Sprint-2
Procurement Officer	Automated Ordering	USN-8	As a procurement officer, I want approved standard orders to be placed automatically with the vendor.	A purchase order is generated and sent to the vendor without manual steps.	High	Sprint-2
		USN-9	As a procurement officer, I can view and download purchase orders and vendor invoices.	Order and invoice documents are available for download.	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
IT Administrator	Catalog Management	USN-10	As an IT administrator, I can add, update or remove standard laptop models in the catalog.	Changes appear immediately in the employee catalog.	Medium	Sprint-2
	Asset Management	USN-11	As an IT administrator, I can confirm delivery and register the laptop in the asset inventory.	The laptop is saved in inventory with an asset tag and owner.	High	Sprint-3
	Reports	USN-12	As an IT administrator, I can view reports on orders, spending and delivery delays.	I can see a dashboard with filters by date and status.	Low	Sprint-3
System Administrator	User & Rule Management	USN-13	As a system administrator, I can manage user roles and approval rules such as budget limits.	Role and rule changes are applied to new requests.	Medium	Sprint-3